Brain states representing Caring and Uncaring Others

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Intro: background and Motivation

- Long term goal: to use neuroscience to overcome the limitations of psychotherapy.
- How people use evidence about others being caring or not has huge consequences for therapeutic relationship.

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- Long term goal: to use neuroscience to overcome the limitations of psychotherapy.
- Using evidence about others being caring or not has huge consequences for therapeutic relationship.
- → apply the neuroscience of **evidence accumulation** to the **interpersonal domain**.
- Value Partitioned Temporal Difference learning as a Bayesian brain approximation (Sands et al, 2024, our pilot work, unpublished)

Hypotheses

- Interpersonal Prediction Errors regarding the salient social states of Caring vs. Not caring, are Value-partitioned and so is their electrophysiological signature.

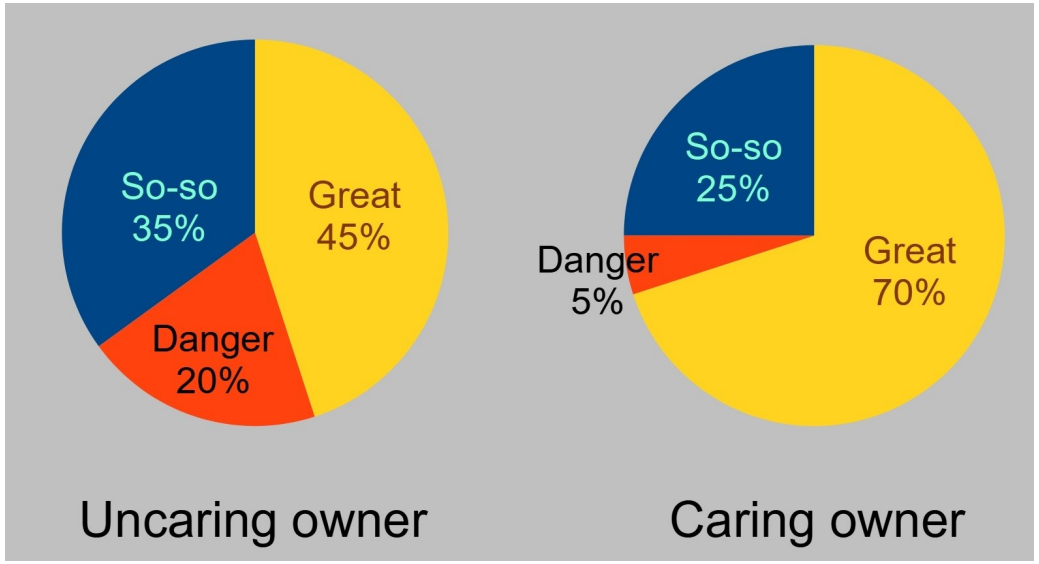
- Caring vs. Uncaring others attract distinct brain representations, involving the ‘social cortex’, especially prefrontal near-midline and temporoparietal cortex.

Methods: Caring Attributions Task

Imagine you are holidaying in faraway towns

The hotel owner gives you advice about where to go for your outing

Some hotel owners are caring, others are uncaring.



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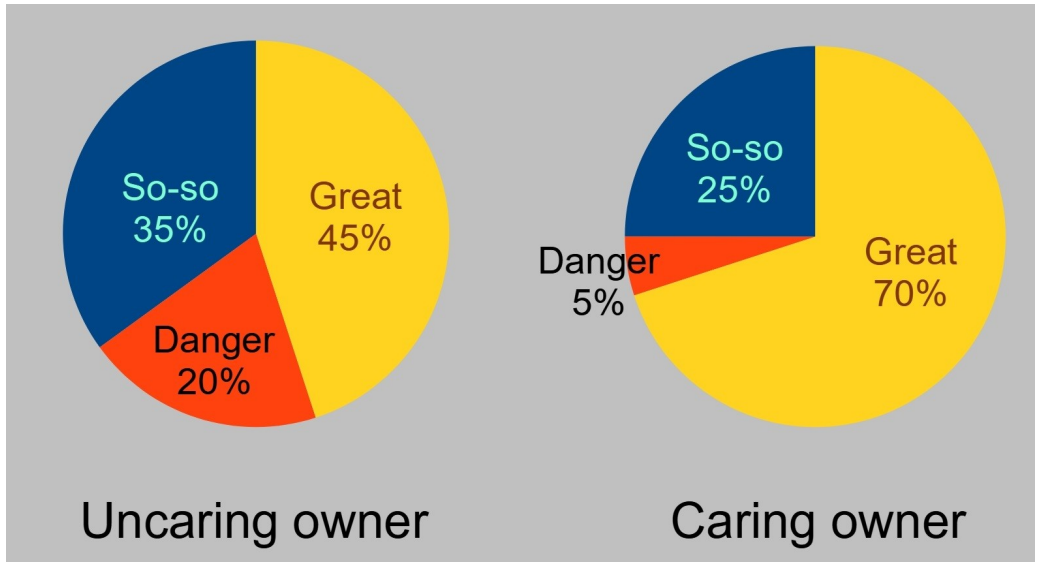
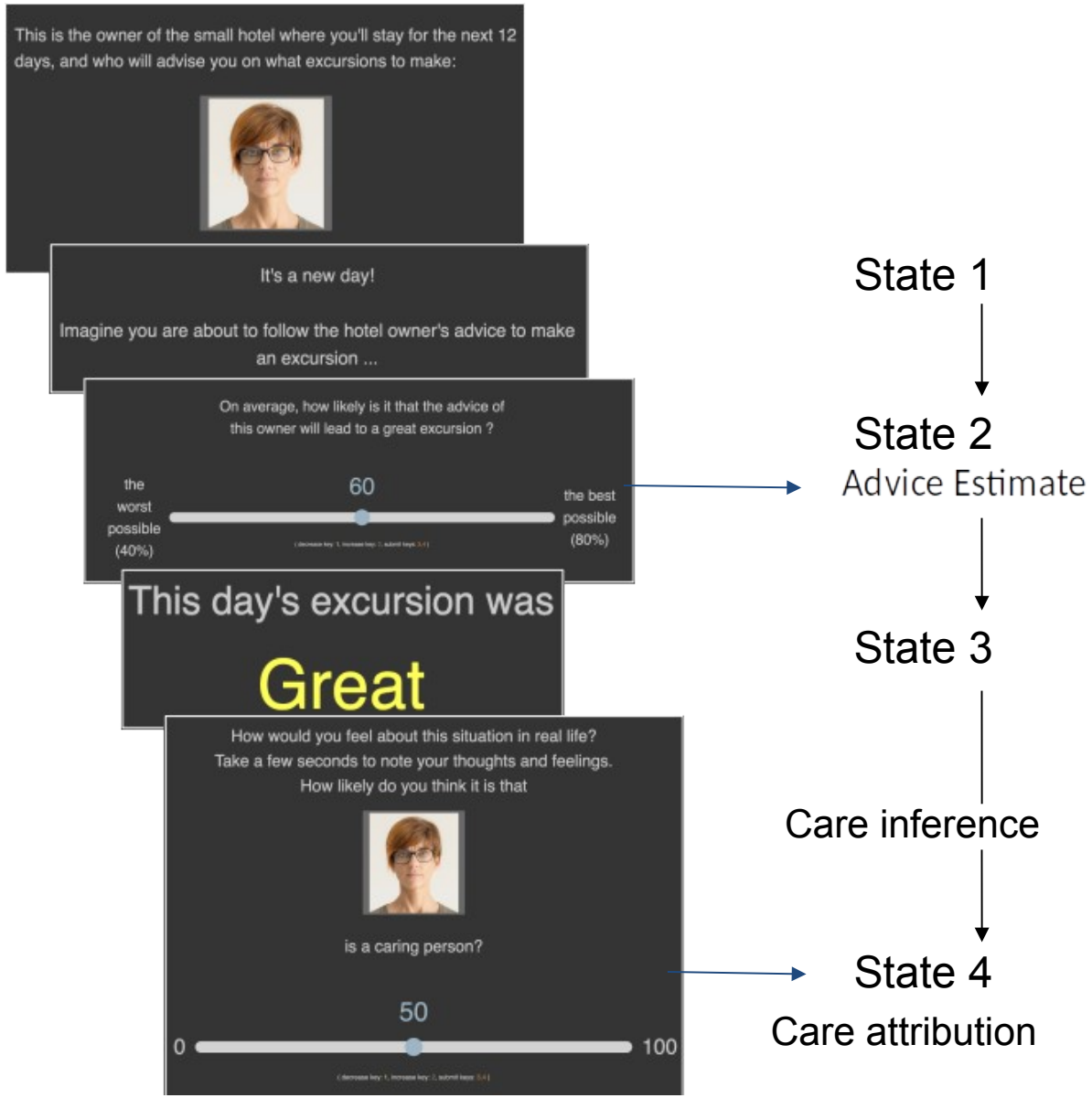
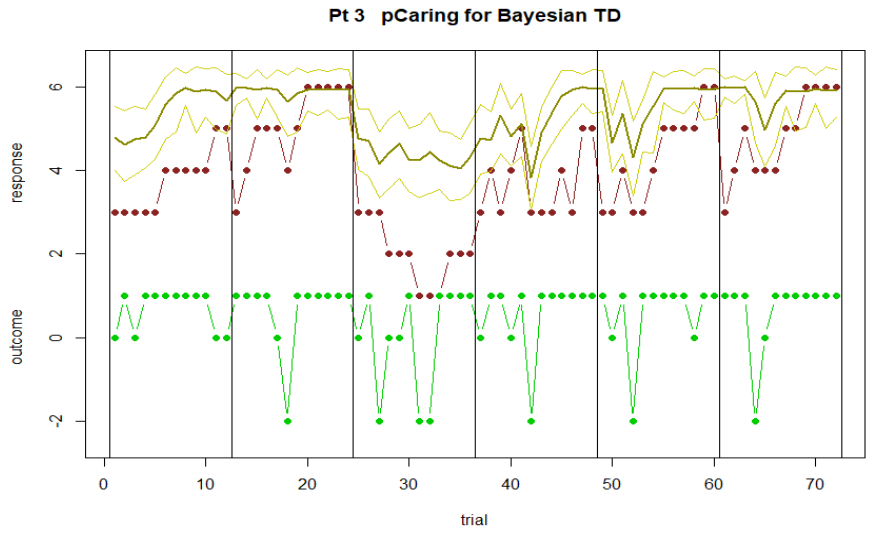




Methods: Analysis

Behavioural

- Associative Temporal-difference Value-partitioned modelling
- Bayesian – Temporal-difference hybrid Value-Partitioned modelling (novel)
- Prediction error parametric regressors for each of four states per trial



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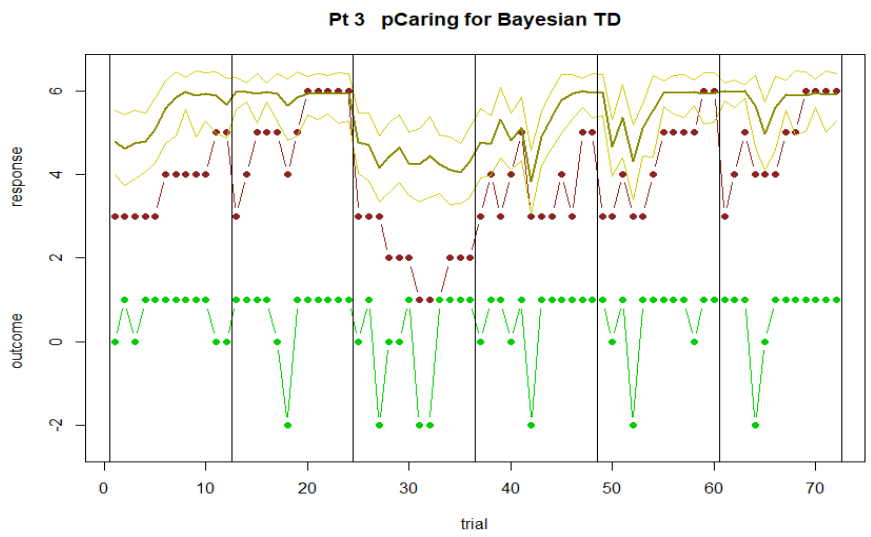

EEG

- Following well established pre-processing (Vladimir Litvak).
- GLM analysis of weighed prediction error estimates as parametric modulator for all four task states (Sands et al, 2024)
- Regression of preprocessed voltage vs. scalp sensors X time interval of interest
- Simple contrast of modeled Uncaring – Caring value over pre-action interval of ‘Care inference’

Logistics - cohort

- How many scanning hours are required per participant?
  - Standard 64 electrode EEG setup x 2 hours / participant
- Are there any cohort considerations?
  - Sensitive discussion of equality and inclusion with respect to culturally important issues e.g. hair.

Logistics – study governance

- What ethics will be used?
  - We have specific ethics for all aspects of the study.
- How is the study funded?
  - Dr Moutoussis discretionary fund including Mr. Frydman Laiter MSc bench fees
- What will be your data sharing policy?
  - As per Ethics permission, anonymized data will be available via Dr. Moutoussis' github within two years of data gathering completion or earlier.
  - Same for analysis code.